

ROTATIONAL MOTION AND ASTROPHYSICS • 1.6
STELLAR PHYSICS

Stellar Physics

How a star spends its whole life
balancing two forces against each
other.

Advanced Higher Physics



Orion: Giuseppe Donatiello, CC0

LEARNING INTENTION

To use the relationships that describe a star's power and brightness, read a Hertzsprung-Russell diagram, and describe how a star forms, lives and ends.

By the end of this topic...

- 1 I can use $P/A = \sigma T^4$, $L = 4\pi r^2 \sigma T^4$ and $b = L/(4\pi d^2)$ to solve problems about stars.
- 2 I can classify a star from its position on a Hertzsprung-Russell diagram, and predict its colour.
- 3 I can describe how a star forms, and state the three stages of the proton-proton chain.
- 4 I can describe how a star leaves the main sequence, and explain what its mass determines.

Daily Review

From Gravitation — close your notes

- 1 State the definition of escape velocity.
- 2 Write the relationship for gravitational potential energy.
- 3 State Newton's universal law of gravitation.
- 4 State the definition of gravitational field strength.
- 5 Sketch the gravitational field around two equal masses.



Every star in this photo is a different colour. What does that tell you?

PART 1

How hot? How big?

Three relationships, and the light already reached us.

By the end of this part...

- 1 I can use $P/A = \sigma T^4$ to find the power radiated per unit area of a star.
- 2 I can use $L = 4\pi r^2 \sigma T^4$ to find a star's luminosity, radius or surface temperature.
- 3 I can use $b = L/(4\pi d^2)$ to find a star's apparent brightness.

Power per unit area

JOTTER NOTES



$$P/A = \sigma T^4$$

P

power (W)

A

surface area
(m²)

σ

a constant
(W m⁻² K⁻⁴)

T

surface
temperature (K)

σ is the Stefan-Boltzmann constant. Every star is treated as a black body.

Power per unit area



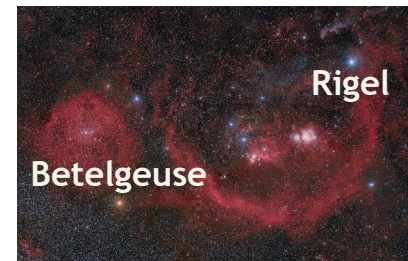
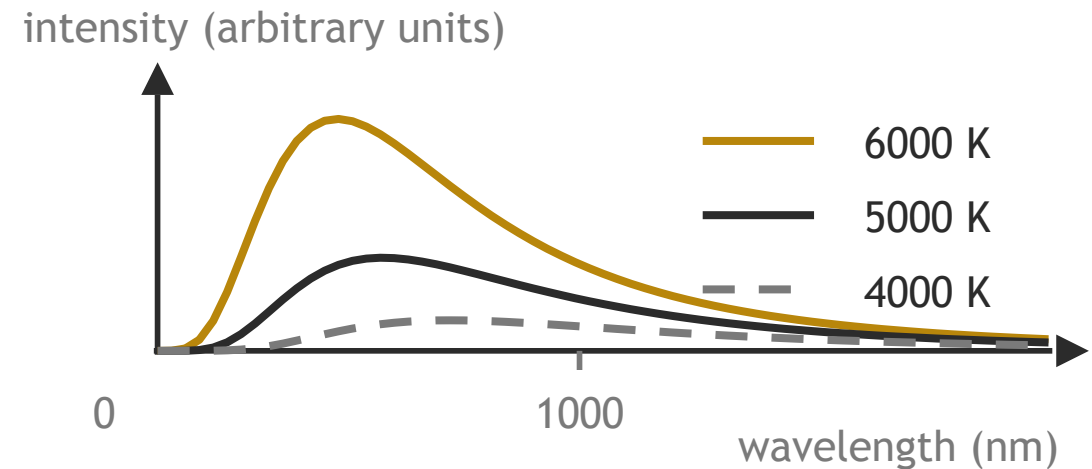
Every star is treated as a black body

$$P/A = \sigma T^4$$

The power radiated per square metre of a star's surface.

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

Double T and P/A rises $\times 16$.



Cool Betelgeuse glows orange; hot Rigel glows blue-white.

Worked example 1 – power per unit area

A star has a surface temperature of 6000 K.

Calculate the power radiated per unit area from its surface.

$$T = 6000 \text{ K}$$

$$P/A = ?$$



$$P/A = \sigma T^4$$

$$P/A = 5.67 \times 10^{-8} \times 6000^4$$

$$\underline{P/A = 7.35 \times 10^7 \text{ W m}^{-2}}$$

Luminosity



$$L = 4\pi r^2 \sigma T^4$$

L

luminosity (W)

r

radius of the
star (m)

σ

a constant
(W m⁻² K⁻⁴)

T

surface
temperature (K)

$4\pi r^2$ is the surface area of the star, so $L = P/A \times A$.

Luminosity

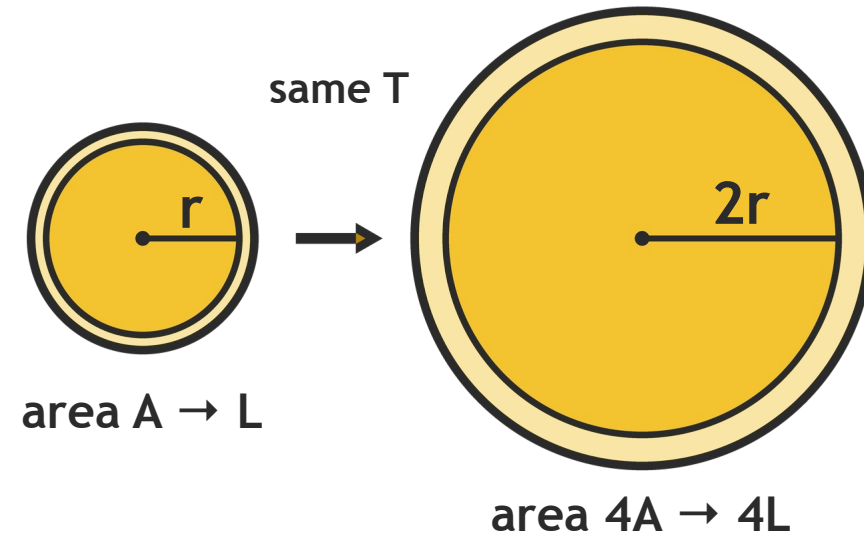


The total power radiated in every direction

$$L = 4\pi r^2 \sigma T^4$$

The TOTAL power radiated, over the star's whole surface area.

A star can be luminous by being hot, by being big, or both.



Worked example 2 – luminosity

A star has a radius of $8.0 \times 10^8 \text{ m}$ and a surface temperature of 5000 K . Calculate its luminosity.

$$r = 8.0 \times 10^8 \text{ m}$$

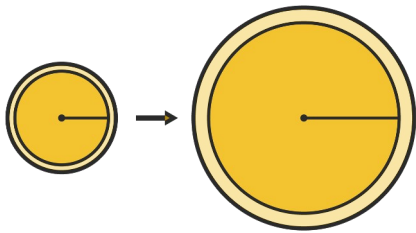
$$T = 5000 \text{ K}$$

$$L = ?$$

$$L = 4\pi r^2 \sigma T^4$$

$$L = 4\pi (8.0 \times 10^8)^2 \times 5.67 \times 10^{-8} \times 5000^4$$

$$\underline{L = 2.85 \times 10^{26} \text{ W}}$$



Apparent brightness



$$b = L/(4\pi d^2)$$

b

apparent
brightness (W m^{-2})

L

luminosity (W)

d

distance to the
star (m)

Convert light-years or parsecs to metres first.

Apparent brightness

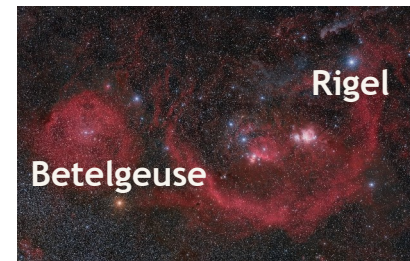
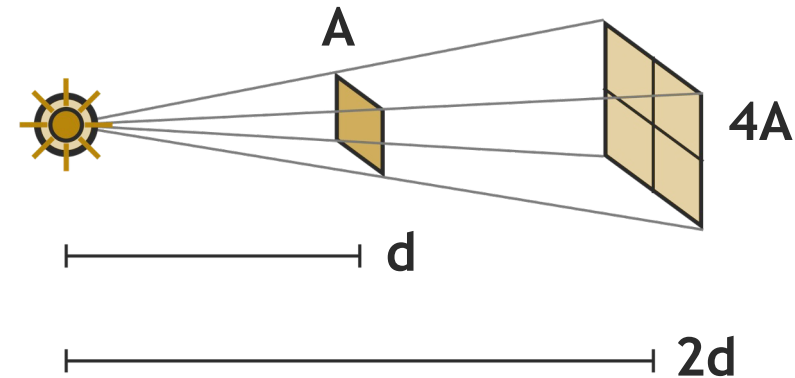


Why a star fades with distance, not just size

$$b = L/(4\pi d^2)$$

The same luminosity, spread over a sphere of radius d .

Two stars can look equally bright for very different reasons – one dim and close, one luminous and far.



From Earth they look alike – but b alone cannot give you L .

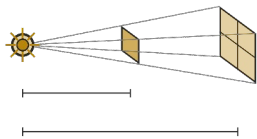
Worked example 3 – apparent brightness – 2023 Q6(d)

Delta Ursae Majoris has a luminosity of $5.46 \times 10^{27} \text{ W}$ and is **24.7 parsecs** from Earth (1 parsec = 3.26 ly). Calculate its apparent brightness.

$$L = 5.46 \times 10^{27} \text{ W}$$

$$d = 24.7 \text{ pc}$$

$$b = ?$$



$$d = 24.7 \times 3.26 = 80.5 \text{ ly}$$

$$d = 80.5 \times 9.46 \times 10^{15} = 7.62 \times 10^{17} \text{ m}$$

$$b = L/(4\pi d^2)$$

$$\underline{b = 7.48 \times 10^{-10} \text{ W m}^{-2}}$$

Daily Review

From Part 1 — close your notes

- 1 Write the relationship for power radiated per unit area.
- 2 Write the relationship for the luminosity of a star.
- 3 Write the relationship for the apparent brightness of a star.
- 4 A star's temperature doubles. By what factor does its power per unit area increase?
- 5 What assumption do all three relationships make about a star?



Astronomers plot clusters like this on a graph. What could a point on that graph tell you?

PART 2

Why plot stars on a graph?

Position says what kind of star it is.

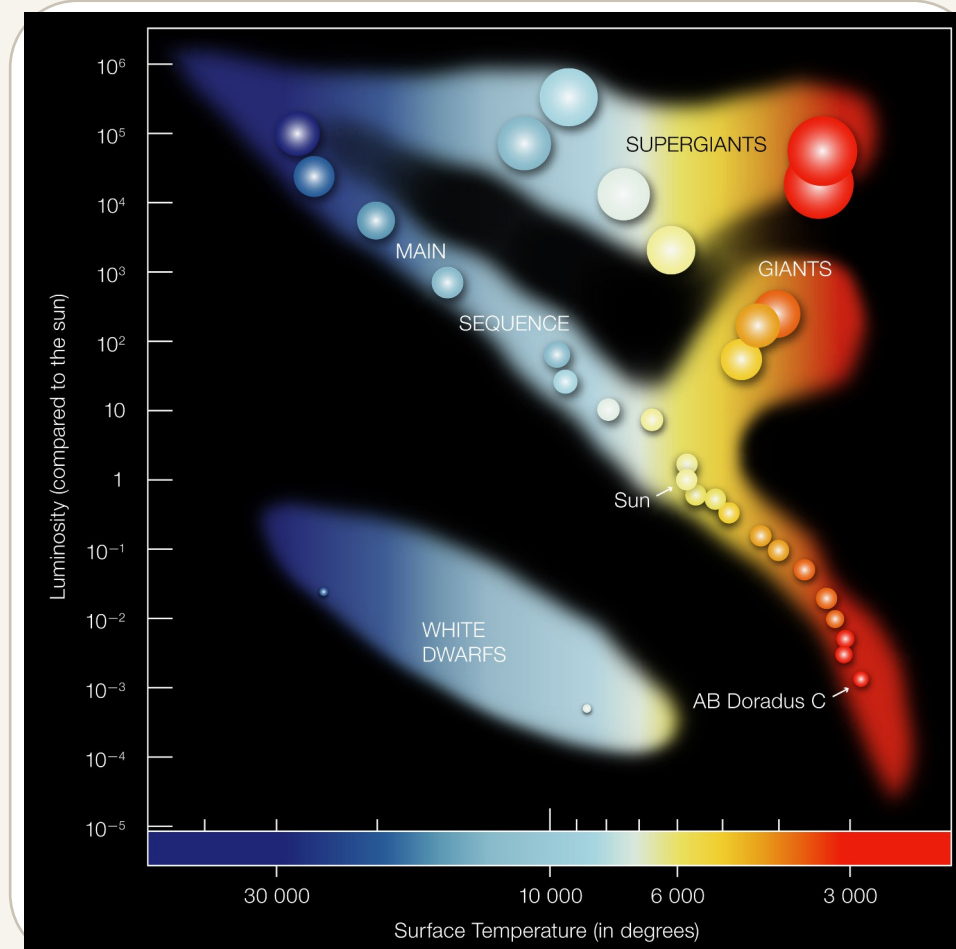
By the end of this part...

- 1 I can describe what a Hertzsprung-Russell diagram represents.
- 2 I can classify a star as main sequence, giant, supergiant or white dwarf from its position.
- 3 I can predict a star's colour from its position on the diagram.

The Hertzsprung-Russell diagram



Temperature runs backwards, and luminosity is a log scale



Reading colour and position

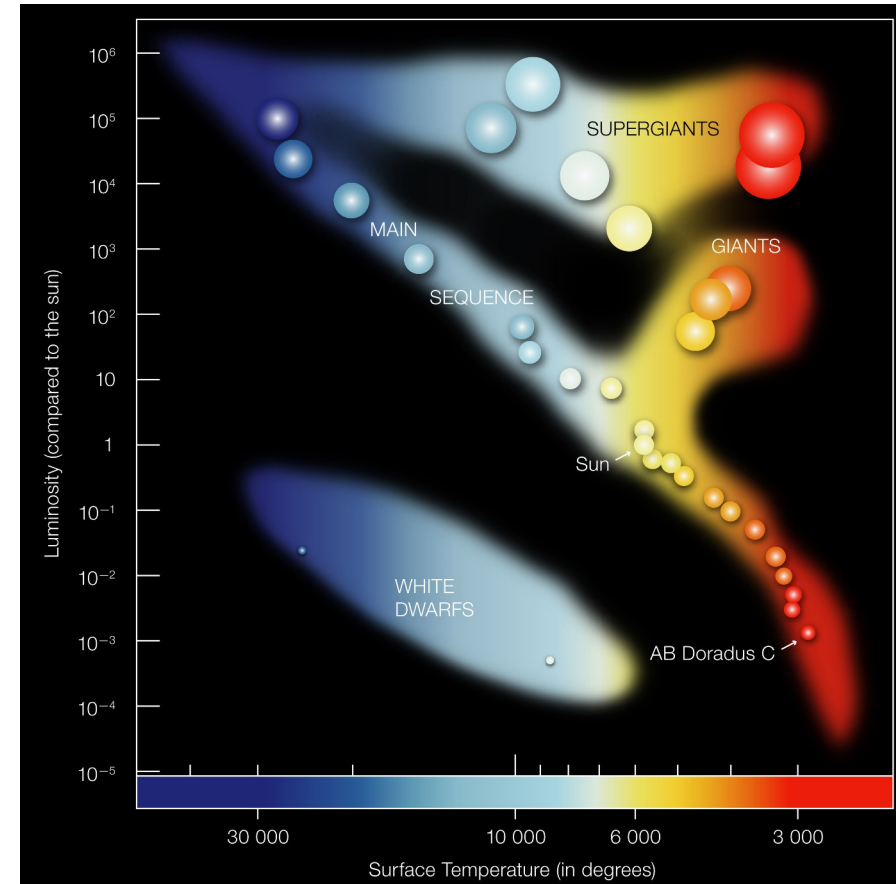


Colour

Hot, blue-white stars sit on the LEFT. Cool, red stars sit on the RIGHT.

Off the main sequence

Giants and supergiants are cool but luminous, so huge. White dwarfs are hot but faint, so tiny.



Worked example 4 – reading the diagram – 2023 Q6(e)

Epsilon Ursae Majoris has a luminosity about two orders of magnitude greater than the Sun's, and a surface temperature of **about 9000 K**. Mark its position on the diagram.

$$L \approx 100 \times L_{\text{Sun}}$$

$$T \approx 9000 \text{ K}$$

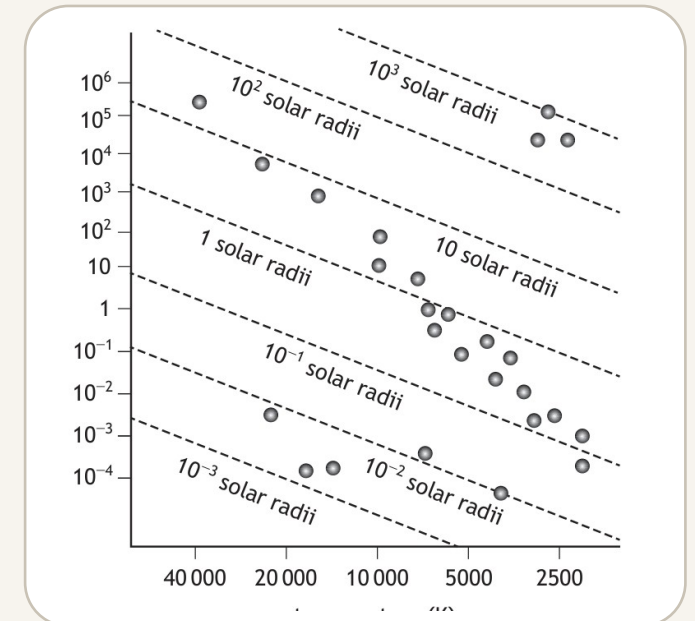
position = ?

2 orders of magnitude = $\times 100$

plot at $L \approx 10^2$, $T \approx 9000 \text{ K}$

just above the main sequence

— in the giant region



Daily Review

From Part 2 — close your notes

- 1 What does a Hertzsprung-Russell diagram plot on each axis?
- 2 Where on the diagram do main-sequence stars lie?
- 3 Where do giants and supergiants lie, compared to the main sequence?
- 4 Where do white dwarfs lie, and why is that surprising given how hot they are?
- 5 What colour is a star with a high surface temperature?



Somewhere in this photo, a star is being born right now. How?

PART 3

What builds a star?

Gravity pulling in, until fusion pushes back.

By the end of this part...

- 1 I can describe how a star forms from an interstellar cloud.
- 2 I can state the three stages of the proton-proton chain.
- 3 I can describe the net effect of the proton-proton chain.

Star formation



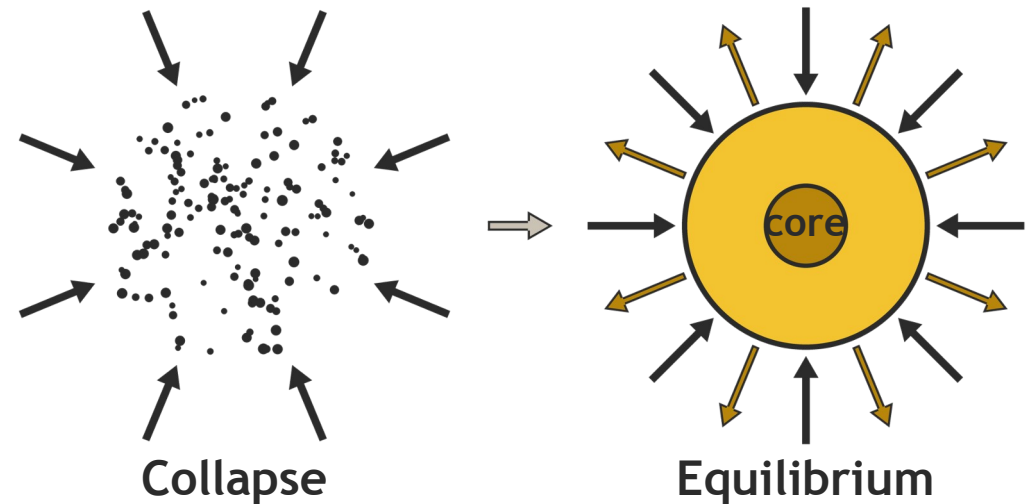
Gravity contracts the cloud until fusion can start

Collapse

Where gravity overcomes an interstellar cloud's thermal pressure, it contracts.

Ignition and balance

The core heats until fusion starts – its thermal pressure then balances gravity.



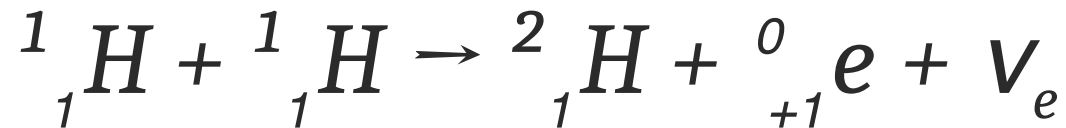
Black arrows: gravity, inward. Gold arrows: thermal pressure, outward.

The proton-proton chain

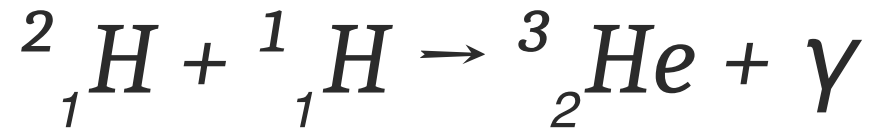


Three stages, hydrogen to helium – spec p.12, verbatim

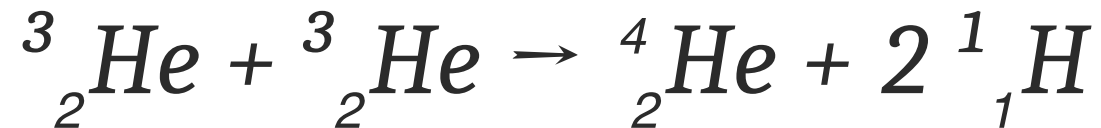
Stage 1



Stage 2



Stage 3



Net effect: 4 protons $\rightarrow$ 1 helium-4 nucleus.

Worked example 5 – the real p-p chain – 2024 Q6(b)

Figure 6A shows one possible p-p chain. State the name of the particles represented by ν_e and ${}^0_{+1}e$, and suggest why deuterium ${}^2_1\text{H}$ does not build up in the star.

$\nu_e = ?$

${}^0_{+1}e = ?$

${}^2_1\text{H}$ build-up = ?

ν_e = (electron) neutrino

${}^0_{+1}e$ = positron

deuterium exists for
only 1 second
before fusing again

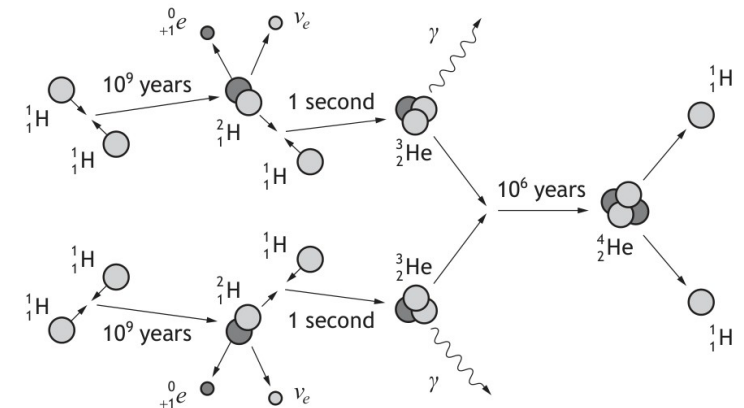


Figure 6A

Daily Review

From Part 3 — close your notes

- 1 What two forces are balanced in a stable, forming star?
- 2 Write the first stage of the proton-proton chain.
- 3 What particles are released at each stage of the proton-proton chain?
- 4 What is the net effect of the proton-proton chain?
- 5 Where in a star does hydrogen fusion happen?



This glowing ring used to be a star. What happened to it?

NASA/JPL-Caltech/ESA, the Hubble Heritage Team STScI/AURA

PART 4

What decides a star's fate?

One number: mass.

By the end of this part...

- 1 I can explain why a main-sequence star is stable.
- 2 I can describe what happens when a star leaves the main sequence.
- 3 I can state that a star's mass determines its lifetime and its eventual fate.

The main sequence



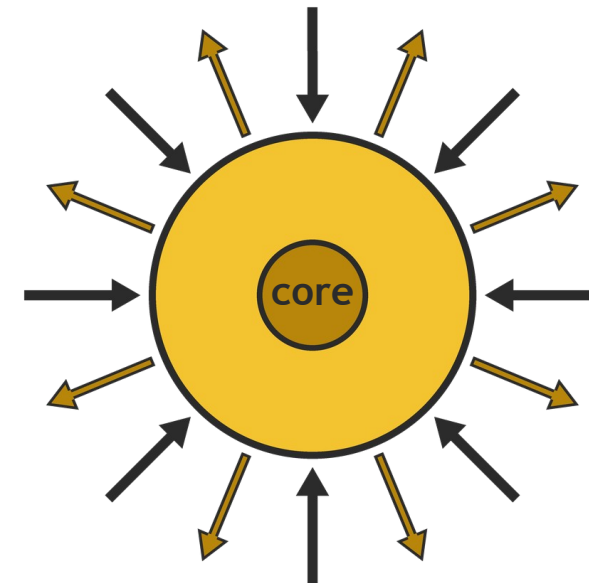
A star spends most of its life here

Balance

Core hydrogen fusion supplies the outward pressure that balances gravity.

The Sun

A main-sequence star, fusing hydrogen in its core right now.



Gravity in (black) balances thermal pressure out (gold).

The Sun — a main-sequence star right now



Prominences on the Sun's surface (SDO).

Leaving the main sequence



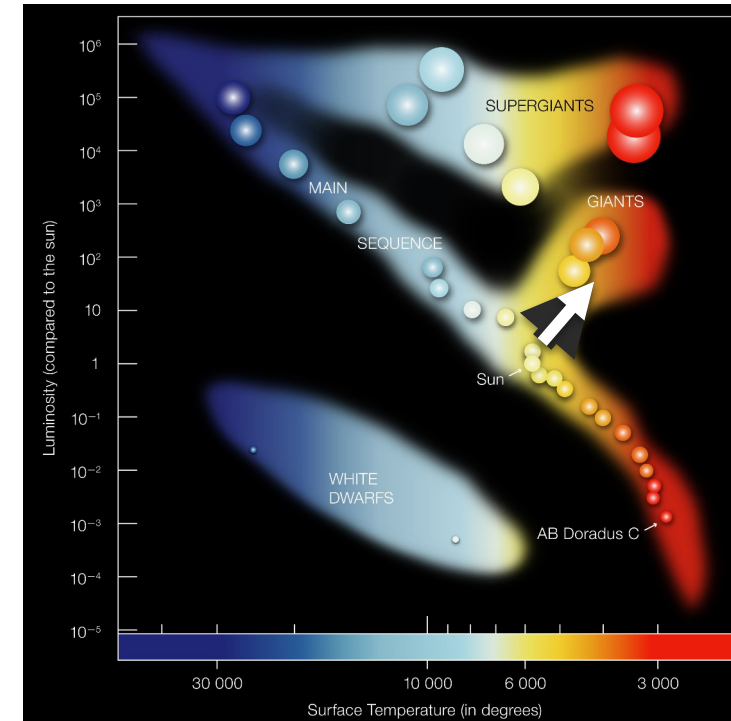
What happens when the core runs out

Core depletes

Hydrogen runs out in the core, but the surrounding shell still has some.

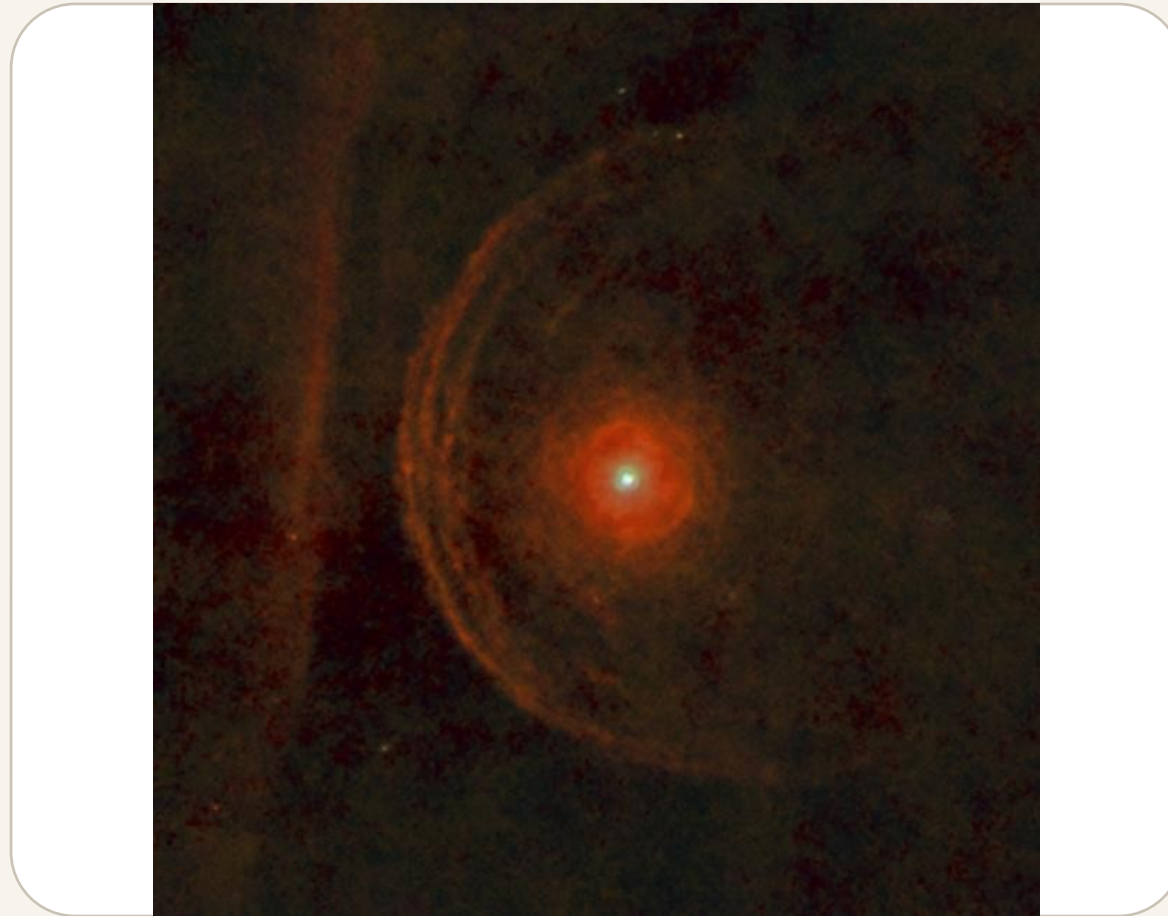
Shell fusion

The shell heats and fuses, pushing the surface outward – the star becomes a giant or supergiant.



Moving up and right, off the main sequence.

Betelgeuse — a real red supergiant

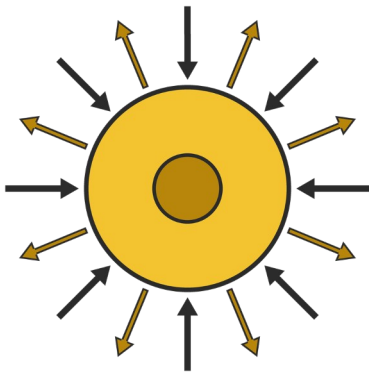


Shells of gas around Betelgeuse (Herschel).

Worked example 6 – becoming a red giant – 2023 Q6(c)

Explain, in terms of thermal pressure and gravitational forces, what happens to the **radius** of the Sun as it becomes a red giant.

radius = ?



radius increases —

thermal pressure > gravity

stops increasing when they balance

Mass decides the fate



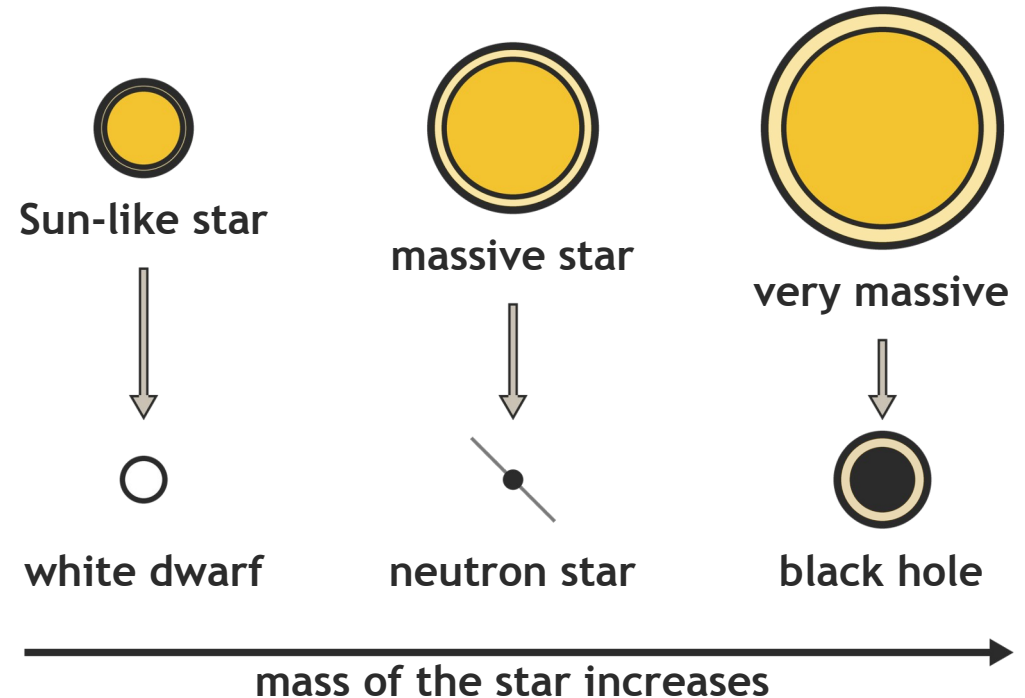
The core keeps contracting and fusing heavier elements

Helium fusion

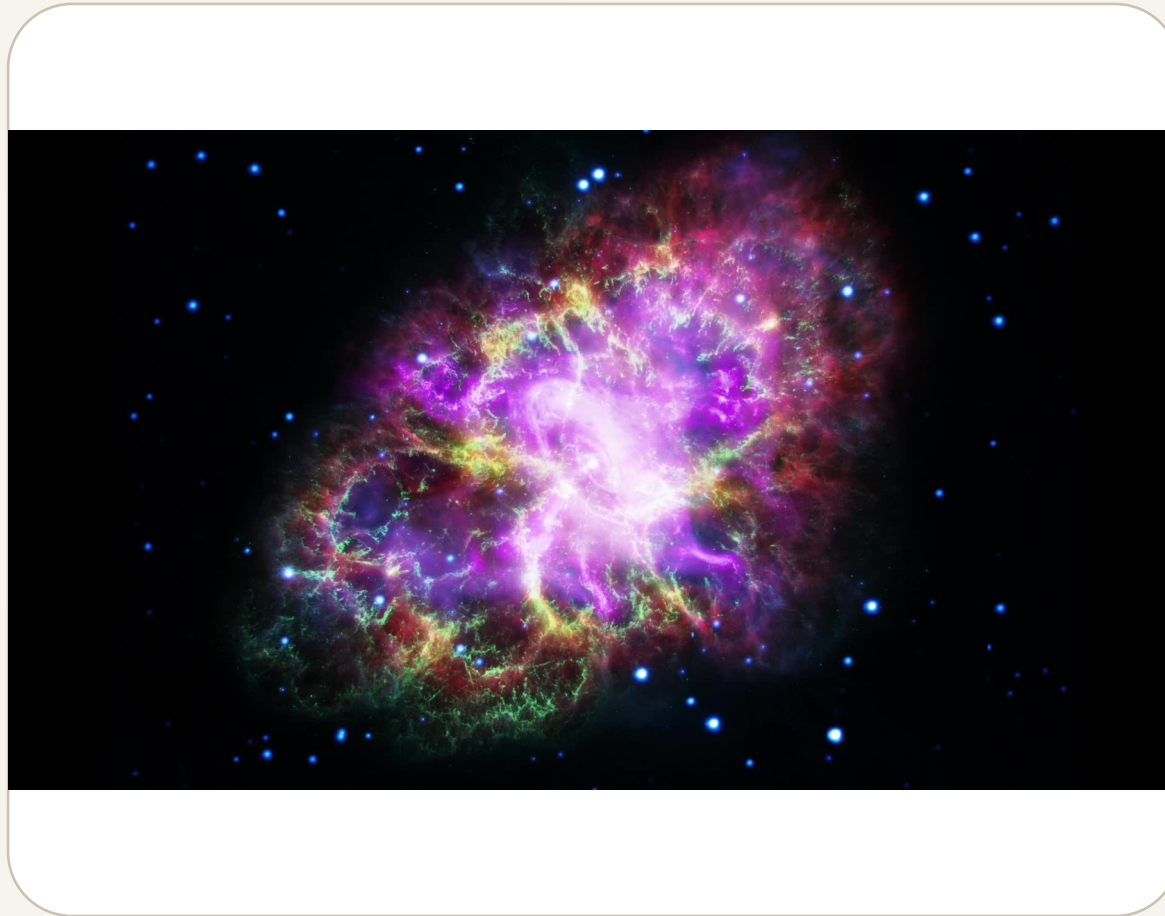
The core shrinks further until it is hot enough to fuse helium.

Three endings

Mass determines lifetime, and the ending: white dwarf, neutron star, or black hole.



The Crab Nebula — a real supernova remnant



A neutron star lies at the heart of this nebula.

Stellar physics in the exam

A question in every diet that sat since 2016

Diet	Question	What it asked
'16,17,19,20,22	varies	Calculation
2018	Q6, Q8	Apparent brightness; p-p chain
2023	Q6	Full topic – worked in class above
2024	Q6	Full topic – worked in class above
2025	Q3, Q5	Evolution; Sirius, Rigel, Betelgeuse

Reading and homework

Scholar: Rotational motion and astrophysics, Topic 7 – Stellar physics

Scholar – Rotational motion, Topic 7 Stellar physics

Scholar – end-of-topic test (homework)

BrightRed Study Guide pp. 46-58

Past papers – 2016-2020, 2022, 2025

Sit the end-of-topic test for homework.



Scholar

Errors that cost marks

The ones that come up every year

- Read carefully which of P/A , L or b is being asked for.
- Convert parsecs and light-years to metres before substituting.
- On an H-R diagram, hot is on the LEFT, not the right.
- A white dwarf is hot but faint — because it is small, not cool.
- Black-hole mechanics is General Relativity, not this topic.

Check yourself

Can you do each of these without your notes?

- 1 Use all three relationships, given any two of L , b , d , r , T .
- 2 Sketch an H-R diagram — main sequence, giants, supergiants, white dwarfs.
- 3 Write all three stages of the proton-proton chain.
- 4 Explain why a star leaves the main sequence for a giant.
- 5 State a star's three possible fates, and what decides between them.

Experiments you could try

Standard kit, or your own phone

- Use a spectrometer to compare colour against temperature.
- Plot stars from a catalogue onto your own H-R diagram.
- Use Stellarium to look up a named star's temperature.
- Model gravitational contraction with a deflating balloon.
- Research a star's class and place it on the H-R diagram.

Course specification – 1 of 7

- Use of appropriate relationships to solve problems relating to luminosity, apparent brightness b , distance between the observer and the star, power per unit area, stellar radius, and stellar surface temperature.
- (Using the assumption that stars behave as black bodies.)

Course specification – 2 of 7

- Knowledge that stars are formed in interstellar clouds when gravitational forces overcome thermal pressure and cause a molecular cloud to contract until the core becomes hot enough to sustain nuclear fusion,
- which then provides a thermal pressure that balances the gravitational force.

Course specification — 3 of 7

- Knowledge of the stages in the proton-proton chain (p-p chain) in stellar fusion reactions which convert hydrogen to helium.
- Knowledge that Hertzsprung-Russell (H-R) diagrams are a representation of the classification of stars.
- Classification of stars and position in Hertzsprung-Russell (H-R) diagrams, including main sequence, giant, supergiant, and white dwarf.

Course specification – 4 of 7

- Use of Hertzsprung-Russell (H-R) diagrams to determine stellar properties, including prediction of colour of stars from their position in an H-R diagram.
- Knowledge that the fusion of hydrogen occurs in the core of stars in the main sequence of a Hertzsprung-Russell (H-R) diagram.
- Knowledge that, in a star like the Sun, the core shrinks and will become hot enough for the helium in the core to begin fusion.

Course specification – 5 of 7

- Knowledge that hydrogen fusion in the core of a star supplies the energy that maintains the star's outward thermal pressure to balance inward gravitational forces. When the hydrogen in the core becomes depleted, nuclear fusion in the core ceases.
- The gas surrounding the core, however, will still contain hydrogen. Gravitational forces cause both the core, and the surrounding shell of hydrogen, to shrink.

Course specification — 6 of 7

- In a star like the Sun, the hydrogen shell becomes hot enough for hydrogen fusion in the shell of the star. This leads to an increase in pressure which pushes the surface of the star outwards, causing it to cool.
- At this stage, the star will be in the giant or supergiant regions of a Hertzsprung-Russell (H-R) diagram.

Course specification – 7 of 7

- Knowledge that the mass of a star determines its lifetime.
- Knowledge that every star ultimately becomes a white dwarf, a neutron star or a black hole. The mass of the star determines its eventual fate.

Sources & credits

- Past papers, Figures 6A/6C: © Qualifications Scotland (SQA).
- SCHOLAR AH Physics notes and tests © Heriot-Watt University.
- Textbook: BrightRed Advanced Higher Physics Study Guide.
- H-R diagram © ESO, CC BY 4.0. Orion: Giuseppe Donatiello, CC0.
- Photos: NASA, NASA/JPL, ESA/Herschel. Other diagrams drawn here.
- © R Stewart Physics, Electronics and Engineering